

Auteur: Thijs Van Schel
Studierichting: Informatica
Oefeningenreeks 4A nr. 62.1

$$\int \sqrt{x^2 - 1} dx$$

$$x = \cosh(t), t = \operatorname{Bgcosh}(x), dx = \sinh t dt$$

$$= \int \sinh^2(t) dt$$

$$= \frac{1}{2} \cosh(t) \sinh(t) - \frac{1}{2} \int dt + C$$

$$= \frac{1}{2} \cosh(t) \sinh(t) - \frac{t}{2} + C$$

$$= \frac{1}{2} x \sqrt{x^2 - 1} - \frac{1}{2} \operatorname{Bgcosh}(x) + C, \text{ want } \sinh(t) = \sqrt{\cosh^2(t) + 1}$$

$$= \frac{1}{2} x \sqrt{x^2 - 1} - \frac{1}{2} \ln |x + \sqrt{x^2 - 1}| + C$$

References